Rapid Convergence Rate in Adaptive Arrays

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Abstract

In many applications, the practical usefulness of adaptive arrays is limited by their convergence rate. The adaptively controlled weights in these systems must change at a rate equal to or greater than the rate of change of the external noise field (e.g., due to scanning in a radar if step scan is not used). This convergence rate problem is most severe in adaptive systems with a large number of degrees of adaptivity and in situations where the eigenvalues of the noise covariance matrix are widely different. A direct method of adaptive weight computation, based on a sample covariance matrix of the noise field, has been found to provide very rapid convergence in all cases, i.e., independent of the eigenvalue distribution. A theory has been developed, based on earlier work by Goodman, which predicts the achievable convergence rate with this technique, and has been verified by simulation.

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I. Introduction

Convergence rate in adaptive systems has been found to be impractically slow with some adaptive implementations. A technique for achieving very rapid convergence in *all* cases is described here. Section II reviews the theory of adaptive radar and presents a new and concise derivation of the optimum weight equation. Section III presents the theory of convergence rate in systems employing the sample covariance matrix for weight computation. These theoretical results were verified by computer simulation, as discussed in Section IV.

II. Optimum and Adaptive Array Processing

Consider an array of M elements. Let $x_k(t)$ be the process, received by the kth element for $K = 1, 2, \cdots, M$. Sample this set of processes at times $t_1, t_2, \cdots, t_L$. For example, for a pulsed radar it is usual to sample periodically at range $R = ct_1/2$, where c is the velocity of light and t_1 is measured from the time of transmission of a pulse. The intervals between consecutive samples, $t_n - t_{n-1}$, are equal to the pulse repetition period. Denote this sampled data set by the column vector

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_N \end{pmatrix}$$

x

where N is the product of M and L; i.e., $N = ML$.

For detection, we choose between two hypotheses, the noise-only hypothesis H_0 and the signal-plus-noise hypothesis H_1. For an array, the signal is the set of voltages $(s_1, s_2, \cdots, s_N)$ an electromagnetic wave from the selected search direction induces on the M receiving elements. The expected value of X, given H_0 (noise only), is

$$EX = 0$$

and the expected value of X, given H_1 (signal plus noise) is

$$EX = S$$

where S is the column vector

$$S = \begin{pmatrix} s_1 \\ s_2 \\ \cdot \\ \cdot \\ \cdot \\ s_N \end{pmatrix}$$

For hypothesis H_1, the "noise" is

$N = X - S.$

To illustrate the signal vector S, let the array be linear with element spacing d. If a plane wave of carrier frequency f impinges on the array at an angle ψ to the array normal, the voltages s_k at the kth element with the carrier removed by heterodyning are

$$s_k = A_k \exp\,[ik(2\pi d/\lambda) \sin \psi + i\delta$$

$$\text{for } k = 1, 2, \cdots, N \tag{1}$$

where $\lambda = c/f$ is the wavelength, δ is a constant phase factor, and c is the velocity of light. A_k for $k = 1, 2, 3, \cdots, N$ is the wave amplitude times a possible amplitude taper, imposed on the elements of the array.

In order to best detect the presence of a signal S, such as (1), one designs a filter which is "tuned to" S in such a manner that the effects of noise and interference are minimized. As we shall see, one criterion which accomplishes this is the maximum signal-to-noise principle. It has been shown elsewhere [1] that if the noise and interference approximate Gaussian processes, then a maximization of the signal-to-noise ratio is equivalent to a maximization of the probability of detection. It is in this sense that the maximum signal-to-noise criterion is an optimal principle.

The filter, needed to detect S, is a weighted sum of the components of X. If the weight vector is

$$W = \begin{pmatrix} w_1 \\ w_2 \\ \cdot \\ \cdot \\ \cdot \\ w_N \end{pmatrix},$$

the output of the filter is the scalar

$$y = \sum_{i=1}^{N} \overline{w}_i x_i = W^* X \tag{2}$$

where $\overline{w_i}$ denotes complex conjugation and W^* denotes the conjugate transpose of the column matrix W. The first- and second-order statistics of y, assuming hypothesis H_1, are the mean

$$Ey = W^* S \tag{3}$$

and the variance

$$\begin{aligned} \mathrm{Var}\,(y) &= E\,|\,y\,|^2 - |\,Ey\,|^2 \\ &= EW^* NN^* W = W^* MW \end{aligned} \tag{4}$$

where M is the covariance matrix, given by

$$M = ENN^* = (En_i \overline{n}_j) \tag{5}$$

for $i, j = 1, 2, \cdots, N$, where the n_j are the noise components of X_j. Clearly, $M^* = M$, so that M is Hermitian.

The noise vector N includes receiver noise of $\psi_0 = kT$ watts per unit bandwidth over bandwidth B for the kth antenna element, where $k = 1, 2, \cdots, N$. Thus, if the sampling times $t_1, t_2, \cdots, t_L$ are also separated sufficiently in time, the covariance matrix has the form

$$M = \psi_0 BI + M_0$$

where I is the $N \times N$ identity matrix and M_0 is the covariance matrix of the external noise only, e.g., sky noise, external interference, etc. Since M_0 is Hermitian and positive semidefinite, there exists a unitary matrix U which diagonalize M_0, i.e.,

$$U^* M_0 U = \Lambda = \mathrm{diag}\,(\lambda_1, \lambda_2, \cdots, \lambda_N)$$

where $\lambda_1, \lambda_2, \cdots, \lambda_N$ are the non-negative eigenvalues of M_0, the covariance matrix of the external noise. Since $U^* = U^{-1}$, the unitary matrix U also diagonalizes M. That is,

$$\begin{aligned} U^* MU &= U^* [\psi_0 BI + M_0]\, U = \psi_0 BI + \Lambda \\ &= \mathrm{diag}\,(\psi_0 B + \lambda_1, \psi_0 B + \lambda_2, \cdots, \psi_0 B + \lambda_N). \end{aligned}$$

Hence, the eigenvalues of M, $\psi_0 B + \lambda_k$, for $k = 1, 2, \cdots, N$ are positive and M is a positive definite Hermitian matrix; i.e.,

$$X^* MX > 0$$

for all vectors $X \neq 0$.

The output signal-to-noise ratio for a complex process associated with narrowband noise is the expected signal power out of filter y divided by the noise power Var (y) given in (4). Thus, by (3) and (4),

$$(S/N)_0 = |\,Ey\,|^2/\mathrm{Var}\,(y) = |\,W^* S\,|^2 / W^* MW \tag{6}$$

where M is the covariance matrix of the noise, (5), and S is the signal vector; e.g., for a linear array the signal would have the form of (1).

The "best" or optimum filter W_0 is obtained by maximizing the signal-to-noise ratio given in (6). Though this maximization is done elsewhere (see [1] and [2]) it is instructive to find W_0 by another method. Here we use the following easily proved fact. If X and Y are two N-component column vectors and M is a positive definite $N \times N$ Hermitian matrix, then

$$(X, Y) = X^* MY \tag{7}$$

is an inner product of the vector space of N-component column vectors. An important property of an inner product is the Schwartz inequality

$$|(X, Y)|^2 \leqslant (X, X)(Y, Y). \tag{8}$$

If the signal-to-noise ratio (6) is re-expressed in terms of the inner product (7), we have

$$\begin{aligned}(S/N)_0 &= |(W, M^{-1}S)|^2/(W, W)\\ &\leqslant [(W, W)(M^{-1}S, M^{-1}S)]/(W, W)\\ &= (M^{-1}S, M^{-1}S) = S^*M^{-1}S\end{aligned} \tag{9}$$

where the bound on the right follows from the Schwartz inequality (8). The bound on the right in (9) can be attained if one lets the filter W in (6) be given by

$$W_0 = kM^{-1}S \tag{10}$$

where k is a complex number. To see this, substitute (10) in (6) as follows:

$$\begin{aligned}(S/N)_0 &= |W^*S|^2/W^*MW\\ &= [1/|k|^2]\left\{[|k|^2|(S^*M^{-1})S\ |^2]/\right.\\ &\quad\left.\cdot\ [(S^*M^{-1})M(M^{-1}S)]\right\}\\ &= (S^*M^{-1}S)^2/S^*M^{-1}S = S^*M^{-1}S.\end{aligned}$$

Thus W_0, as given by (10), is the optimum filter, the optimum set of filter weights, for the detection of S in the presence of noise M.

In practice, one generally does not know a priori the clutter and interference situation. When it is known and an interference model can be constructed, the "best" set of antenna weights is computed in accordance with W_0, as given by (10). However, if the interference field changes, for example, by the presence of moving near-field scatterers, antenna errors, interference, and jamming, one must continually update or *adapt* the weight vector W_0 to meet the varying conditions.

The weights, the components of W_0, are adapted to new situations by processing a number of independent samples of received data. The different methods of processing the data to estimate W_0 require different numbers of independent samples for the same level of performance. That is, the convergence to achieve a certain signal-to-noise ratio is dependent on the adaption criterion. The purpose of this paper is to compare the convergence rates of the already familiar Widrow and Applebaum recursive algorithms (see [1] and [2] for the development of these algorithms) with the method of direct sample matrix inversion of M, where M is the estimate of the covariance matrix M using K independent samples of data. Both computed and theoretical results will be used to make this comparison. Finally, some discussion of the trade-offs between equipment complexity and the rate of convergence will be given.

III. Rapid Adaptivity By Sample Matrix Inversion

In this section we show how the adaptive array-filter processor can be achieved by first estimating the covariance matrix M, using K samples. Next, this estimate $\hat{M}$ of M is inverted to form, finally, the filter $k\hat{M}^{-1}S$. In the next section, this method of adaptivity will be compared by example with the LMS (least-mean-square) algorithm of Widrow [2] and the MSN (maximum signal-to-noise)[1] algorithm of Applebaum [1].

Suppose the noise process impinging on the receiving elements is observed through the N-dimensional vector X for K *independent* samples. In a range sampling radar, for example, these samples might be the returns from K different and independent range intervals. Let $X^{(j)}$ for $j = 1, 2, 3, \cdots, K$ denote the jth sample of vector X. For noise only, $X^{(1)}, X^{(2)}, \cdots, X^{(K)}$ are K independent and identically distributed N-variate complex vectors. Each of these vectors is distributed with probability density

$$P(X) = (\pi)^{-N}|M|^{-1}\exp(-X^*M^{-1}X)$$

where $|M|$ is the determinant of covariance matrix M [3], [4]. Thus, the joint probability density of $X^{(1)}, X^{(2)}, \cdots, X^{(K)}$ is

$$P(X^{(1)}, X^{(2)}, \cdots, X^{(K)}) = (\pi)^{-NK}|M|^{k}\cdot\exp-\left(\sum_{i=1}^{K} X^{(i)*}M^{-1}X^{(i)}\right) \tag{11}$$

where M is the covariance or complex moment matrix, (5),

$$M = EXX^*$$

where X is the sampled-data vector for noise alone.

In order to obtain the estimate $\hat{M}$ to substitute for M in (10) of the optimum filter W_0, one uses the maximum-likelihood principle. The maximum-likelihood estimate of M has been shown by Goodman [3, Theorem 4.1] to be

$$\begin{aligned}\hat{M} &= 1/K\sum_{j=1}^{K} X^{(j)}X^{(j)*}\\ &= 1/K\sum_{j=1}^{K}(x_r^{(j)}\overline{x}_s^{(j)})\end{aligned} \tag{12}$$

where $x_r^{(j)}\overline{x}_s^{(j)}$ denotes the elements in the rth row and sth column of matrix $X^{(j)}X^{(j)*}$. This result is obtained by maximizing the logarithm of the likelihood function

$$L = P(X^{(1)}, X^{(2)}, \cdots, X^{(K)}),$$

as given by (11). $\hat{M}$, as given by (12), is simply the arithmetic average of the K sample matrixes $X^{(j)}X^{(j)*}$ for $j = 1, 2, \cdots, K$. $\hat{M}$ is called the sample covariance matrix.

[1] In general, this algorithm maximizes the ratio of signal to noise, where the noise term includes both receiver noise and all external noise components.

If one chooses $\hat{M}$ for M in the optimum matched filter criterion (10), the filter has the form

$$\hat{W} = k\hat{M}^{-1}S \tag{13}$$

where $\hat{M}$ is given by (12). Suppose $\hat{W}$ as given above is used to filter and detect the presence of a signal S in another independent sample of vector X. By (2), substituting $\hat{W}$ for $\hat{W}$, the output of this filter is

$$\hat{y} = \hat{W}^* X. \tag{14}$$

Let us now compute the output signal-to-noise ratio $(S/N \mid \hat{W})_0$, assuming that the estimated filter $\hat{W}$ is held fixed.

The first- and second-order statistics of $\hat{y}$, holding $\hat{W}$ fixed, i.e., the conditional first and second statistics conditioned on $\hat{W}$, are computed as follows:

$$E(\hat{y} \mid \hat{W}) = \hat{W}^* S$$

and

$$\begin{aligned}\mathrm{Var}\,(\hat{y} \mid \hat{W}) &= E(\mid \hat{y} \mid^2 \mid \hat{W}) - \mid E(\hat{y} \mid \hat{W}) \mid^2 \\ &= \hat{W}^* \,[E(X-S)(X-S)^*]\, \hat{W} \\ &= \hat{W}^* M \hat{W} = S^* (\hat{M})^{-1} M (\hat{M})^{-1} S.\end{aligned}$$

Thus, by (6), the output signal-to-noise ratio, conditioned on $\hat{W}$, is

$$\begin{aligned}(S/N \mid \hat{W})_0 &= [E(\hat{y} \mid \hat{W})]^2 / \mathrm{Var}\,(\hat{y} \mid \hat{W}) \\ &= (S^* \hat{M}^{-1} S)^2 / S^* \hat{M}^{-1} M \hat{M}^{-1} S.\end{aligned} \tag{15}$$

Since $\hat{M}$, by (12), is a function of the random data vector $X^{(j)}$ for $j = 1, 2, \cdots, K$, the above-conditioned signal-to-noise ratio in (15) is a random variable. It is convenient in dealing with this random variable to normalize it with respect to its upper bound, namely, the right side of (9). Denote this normalized signal-to-noise ratio by $\rho(\hat{M})$. That is, let

$$\begin{aligned}\rho(\hat{\mathrm{M}}) &= (S/N \mid \hat{W})_0 / S^* M^{-1} S \\ &= (S^* \hat{M}^{-1} S)^2 / [(S^* M^{-1} S)(S^* \hat{M}^{-1} M \hat{M}^{-1} S)].\end{aligned} \tag{16}$$

Evidently, the random variable ρ lies in the interval $0 \leqslant \rho(\hat{M}) \leqslant 1$.

In [4], Capon and Goodman used the Goodman theory [3] of the complex Wishart distribution to find the probability distribution of certain estimators, similar to the random variable $\rho(\hat{M})$ defined above. Also, Goodman, in unpublished notes, found that the distribution of the inverse of $\rho(\hat{M})$ satisfied an incomplete beta function distribution. By a simple change of variable, the probability density of the normalized signal-to-noise ratio $\rho(\hat{M})$ is likewise a beta function distribution, namely,

$$\begin{aligned}P(\rho) = &\left\{K!/[(N-2)!(K+1-N)!]\right\} \\ &\cdot (1-\rho)^{N-2} \rho^{K+1-N}\end{aligned} \tag{17}$$

where $0 \leqslant \rho \leqslant 1$ and $K \geqslant N$. An outline of the derivation of (17) is included in the Appendix.

The expected value of $\rho(\hat{M})$ can be computed, using (17), to be

$$E(\rho(\hat{M})) = (K + 2 - N)/(K + 1). \tag{18}$$

This is the expected loss in power ratio if only K samples of data were used to estimate $\hat{M}$ in (12). Expressed in decibels, this expected loss in power ratio is

$$\begin{aligned}\text{loss} &= -10 \log_{10} \left\{E[\rho(\hat{M})]\right\} \\ &= -10 \log_{10} [(K + 2 - N)/(K + 1)].\end{aligned} \tag{19}$$

If one wishes to maintain an average loss ratio of better than one-half (less than 3 dB) by (17) and (10), at least $K = 2N - 3 \cong 2N$ samples of data are needed. This useful "rule-of-thumb" requirement on the number of data samples will be shown in the next section to agree with results obtained by simulation. In general, the loss in decibels compared to the optimum, given in (19), supplies the trade-off between performance and the number of samples needed for filter adaptation in the method of sample matrix inversion (SMI).

IV. Verification of Theoretical Results

A. Comparison of Adaptive Criteria by Simulation

By (18), if the number of samples K gets large, the expected value of $\rho(\hat{M})$ approaches one. Thus, $\hat{M}$, the sample covariance matrix, converges to the covariance matrix M, given in (5). When $\hat{M}$ is generated by computer simulation, using a set of fixed clutter points with a different random phase and amplitude applied to each point of a sample, it likewise will approach M when the number of samples K is large.

The question of how many samples are required to approach the covariance matrix $\hat{M} \rightarrow M$ was addressed using a computer simulation. The simulation was of a moving platform radar with four antenna elements and two sampling pulses ($N = 8$ complex inputs). The interference clutter consisted of 30 scatterers. These scatterers were spaced equally on a semicircle of large radius centered on the array. The array lying on the semicircle diameter was imagined to move a distance of 0.2 wavelengths along its length between pulses to simulate platform motion. The scatterers were illuminated on each pulse by a broadside

TABLE I

4 Elements; 2 Pulses [$N = 2 \times 4$]; Spacing = 0.5 λ
Interpulse motion = 0.2 λ/pulses; Steady state gain = 62.2 dB

No. of Samples K	SMI Theory*	SMI Simulation (MSN)	Adaptive Loops (MSN)
5		28.4	15.3
10		55.6	15.5
15	59.7	58.9	15.8
20	61.0	58.6	17
800			37
2000			38
∞	62.2		

*Values given by 62.2 ± 10 $\log_{10}$ [$(K - N + 2)/(K$ ¼ $1)$].

transmitter beam formed by the four equally weighted antenna elements.

The ideal or steady-state covariance matrix M was first computed by making use of the known positions of clutter and elements to sum the covariance matrixes for each clutter point. This yields the steady-state or ideal matrix M, where each matrix element M_{mn} is

$$\begin{aligned} M_{mn} &= E\left\{V_m \overline{V}_n\right\} = E\left\{\Sigma V_{mj} \Sigma \overline{V}_{nj}\right\} + R_m \delta_{mn} \\ &= E\left\{\sum_{j=1}^{J} V_{mj} \overline{V}_{nj} + \left[\sum_{k=1}^{J} V_{mk} \sum_{\substack{l=1 \\ l \neq k}}^{J} \overline{V}_{nl}\right]\right\} + R_m \delta_{mn} \\ &= \sum_{j=1}^{J} V_{mj} \overline{V}_{nj} + R_m \delta_{mn} \end{aligned}$$

where

$E\{\ \}$ = expected value
$\overline{V}_{mj}$ = conjugate voltage at mth antenna element from jth clutter point; m and n range over all antenna elements for each pulse
$\delta_{mn} = 1, m = n$
$= 0, m \neq n$
R_m = average value of receiver noise at each element on each pulse.

From this, the optimum filter weights were obtained from (10), using matrix inversion. These weights, in turn, were then used to compute the signal-to-clutter ratio [the signal-to-noise ratio given in (6)]:

$$S/C = | W^* S |^2 / W^* M W. \tag{20}$$

The MTI gain is defined as the S/C ratio, given by (20), divided by a normalizing constant, the ratio of signal-to-clutter returned on one pulse, assuming uniform antenna weights. The steady-state maximum MTI gain for the above problem was computed to be 62.2 dB. A sample covariance matrix $\hat{M}$ was then computed using different numbers of samples k, where

$$\hat{M}_{mn} = \sum_{i=1}^{K} V_{mi} \overline{V}_{ni}. \tag{21}$$

For each sample i, every clutter point is given a random phase and amplitude and the complex voltages are added at each element for each pulse. Weights were obtained by inverting the sample covariance matrix $\hat{M}$, as in (13), and these weights were used to compute the MTI gain, using (20), where M is the steady-state expected value covariance matrix. Some results are shown in column 3 of Table I. The simulated results in column 3 show the rapid convergence to the steady-state value of 62.2 dB and the close comparison with theoretical prediction column 2 for a single run. The experiment was performed several times with different random numbers for determining clutter phase and amplitude, with similar results.

For comparison, some results are shown in column 4 of Table I, using the maximum signal-to-noise (MSN) loops of Applebaum [1]. More results for the MSN algorithm are shown in Fig. 1. The MSN algorithm and least-mean-square (LMS) algorithm of Widrow [2] both derive the next set of weights from the previous set using a recursive relation, which, in digital form, can be expressed as

$$W_i(j+1) = W_i(j)[1 - (1/\tau)] + (g/\tau)e(j)V_i^*(j) - (S_i^*/\tau)$$

where

$$e(j) = P(j) - \sum_{i=1}^{N} W_i(j) V_i(j)$$

is the difference between the received signal at the output and the transmitted pilot signal $P(j)$ (present only in the LMS case), and where g is the loop gain, τ is a filter smoothing constant, and S_i^* are the steering vectors. For LMS, $S_i^* = 0$. For MSN, $S_i^* \neq 0$ and $P(j) = 0$. The MSN algorithm usually converges more rapidly because the steering vectors are supplied. These algorithms require essentially only one complex multiply per sample for each channel, i.e., $g/\tau e(j)V_i^*(j)$, and are therefore simple compared to the direct inversion method.

Both algorithms yield optimum steady-state weights and maximum MTI gain in the limit. The number of samples required to approach the limit using these approximations

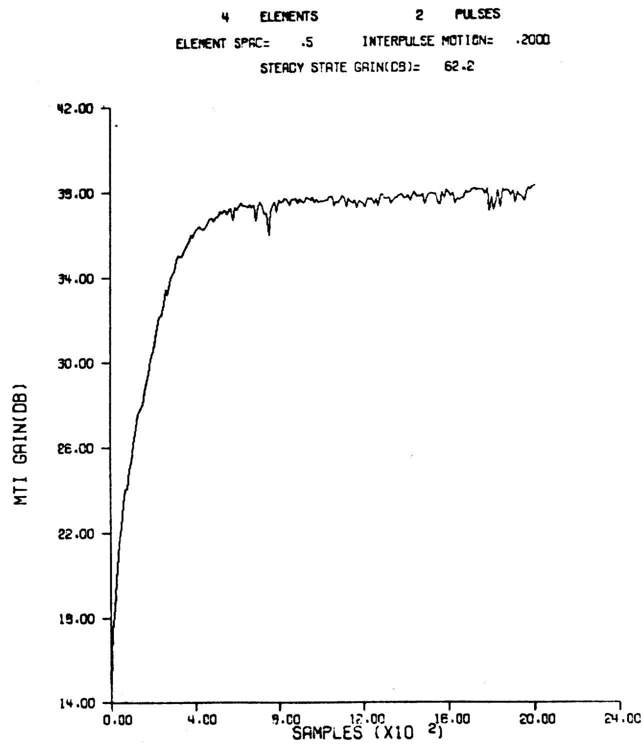

Fig. 1. Simulation of MSN algorithm.

may be very large—10^9 or more samples—and there are frequently plateaus, similar to that shown in Fig. 1 around 38 dB, that are significantly below the optimum performance of, in this case, 62 dB.

The slow convergence of the MSN and LMS loops can be understood in terms of the eigenvalues of the noise covariance matrix. A solution for the transient response of these algorithms has been found by transforming the vector equations into normal coordinates [1]. The time response is then given by a sum of exponentials in which the time constants are a function of the eigenvalues of the covariance matrix. When the eigenvalues differ by orders of magnitude, the performance converges to different levels in widely different numbers of samples.

Convergence using these algorithms is highly example-dependent. In some cases it is very fast. This usually occurs when a discrete set of noise sources is present and the number is less than the number of adaptive loops. When interference is widely distributed, as in this example, there often exists a good solution, but the number of samples required to attain it can be very high. Using the SMI method, convergence is independent of the noise environment and depends only on the number of adaptive elements.

B. SMI With Signal Present

Adaptive systems do not converge fast when weights are formed with signal present. In radar applications, it is usually possible to form the interference covariance matrix with signal absent. To determine the effect of having some signal present, experiments were run with signals of varied amplitude present in one of the samples used to form the sample covariance matrix. The weights determined from this covariance matrix were used with the expected value covariance matrix to determine MTI gain. Some results of this experiment are shown in Table II.

The small degradation in MTI gain and in the output S/C can be seen as a function of input clutter-to-signal ration. When large signals are present, the output S/C ratio is degraded, but appears to be enough for detection.

C. Estimating Relative Complexity

An advantage of the MSN algorithm is the simplicity of its implementation, especially when analog techniques are employed. The shift to digital processing, however, makes more sophisticated processing possible, and preferable in some cases. Not only can performance be achieved that was not practical before, but in many cases it can be done more cheaply. While the MSN algorithm is very simple to implement conceptually, its convergence rate can be very slow, as discussed earlier. The SMI technique, on the other hand, converges with the smallest possible number of samples, but requires more processing steps, and usually more accuracy, i.e., bits. To compare these techniques when digital processing is used, we can compare the number of complex multiplications and the amount of storage required as a way of estimating cost.

TABLE II

4 Elements; 2 Pulses; Spacing = 0.5 λ
Interpulse motion = 0.2 λ/pulse; Steady-state gain = 62.2 dB

Signal Present in One Sample						
C/S (dB) Input	10 Samples MTI Gain	10 Samples S/C Out	20 Samples MTI Gain	20 Samples S/C Out	∞ Samples MTI Gain	∞ Samples S/C Out
75	55.6	-19.4	58.6	-16.4	62.2	-13
55	57.9	2.9	59.6	4.6	62.2	7
35	44.4	9.4	56.8	21.8	62.2	27

Usually when loops are used, all the incoming data is processed. This may not be practical or desirable, however, when using a sample covariance matrix. In the latter case, advantage can be taken of the small number of samples required and the fact that new weights are required only as often as there is change in position or distribution of the interference. These changes are often of the order of 0.01 to 1 seconds, rather than microseconds, which reduces the cost of SMI processing. Spatial interferers, for example, do not usually change angular direction rapidly and if several at different locations blink on and off, both will be nulled so long as the sampling interval is either long or short compared to the blink rate. Motion of the antenna itself causes a fast change, but this can sometimes be avoided by using electronic scan.

Formation of the sample covariance matrix requires $SN(N+1)/2$ complex multiplications, where S is the number of samples required for convergence and N is the total number of weights, i.e., antenna elements times pulses for a space time processor. To invert the Hermitian matrix requires $(N^3/2 + N^2)$ complex multiplications, and to form each set of weights requires another N^2 multiplications. If only one set of weights is required, i.e., no filter banks or multiple beams are needed, then the weights can be found by solving the set of equations $W = \hat{M}^{-1}S$, which requires about $N^3/6$ complex multiplications. Substituting $S = 2N$ for the number of samples required to be within 3 dB of optimum, the total number of complex multiplications required to obtain a set of weights is approximately $N^3 + N^3/6 = 7/6N^3$ for SMI, compared with NS for the MSC algorithm, where S can be very large. The rate of multiplication, a measure of processing cost, is the number of multiplications divided by the time interval between required weight changes. Often the multiplication rates can be quite low for very good performance.

In addition to the multiplication rate, the accuracy or number of bits required must be considered in estimating complexity. The MSN algorithm requires very low multiplication accuracy, while SMI requires a high accuracy in forming the sample matrix and solving the equations for the weights.

D. Conclusion

Forming a sample covariance matrix and solving for the weights provides a very fast rate of convergence. This rate is dependent only on the number of weights and is independent of the noise and interference environment. It has been found to be orders of magnitude faster than the MSN or LMS algorithms for a number of practical problems. Because of the small number of samples required and the often slow rate of required weight change, it may be economical to implement in many practical radar, communications, and sonar applications.

Appendix

Distribution of $\rho(\hat{M})$

The purpose of this appendix is to outline the method for finding the distribution, (17), of the normalized conditional output signal-to-noise ratio $\rho(\hat{M})$ given in (16). This development is based in part on a paper of Capon and Goodman [4], and on unpublished material made available to one of the authors by N.R. Goodman.

In [3], Goodman shows that the joint distribution of the elements of the estimated covariance matrix (12)

$$\hat{M} = 1/K \sum_{j=1}^{K} x^{(j)} x^{(j)*}$$

is given by the central complex Wishart distribution

$$P(A) = (|A|^{K-N}/I(M)) \exp\left[-tr(M^{-1}A)\right] \tag{22}$$

where $A = K\hat{M}$ tr(X) denotes the trace of X, i.e.,

$$\text{tr } X = \text{tr } (x_{ij}) = \sum_{i=1}^{N} x_{ii},$$

M denotes the covariance matrix (5), and $I(M)$ denotes the constant

$$I(M) = \pi^{(1/2)N(N-1)}\Gamma(K)\Gamma(K-1)$$
$$\cdots \Gamma(K-N+1)|M|^K. \tag{23}$$

The volume element, associated with density (22), is

$$dV = dA_{11} dA_{22} \cdots dA_{NN} dR(A_{12}) dI(A_{12})$$
$$\cdot\ dR(A_{13}) dI(A_{13}) \cdots dR(A_{N-1,N}) dI(A_{N-1,N})$$

where $R(A_{ij})$ and $I(A_{ij})$ denote the real and imaginary parts of A_{ij}, respectively. The complex Wishart density is defined only over a domain D_A of $N \times N$ matrixes A for which A is positive definite. To understand the complex Wishart distribution of Goodman, it is helpful to compare its development with that of the real Wishart distribution, given by Cramer [5].

The fundamental parameters of the central complex Wishart distribution (22) are K, the sample size, N, the number of dimensions of vector X, and M the $N \times N$ covariance matrix of X under the noise-only hypothesis H_0. Thus, following Capon and Goodman [4], one designates a central complex Wishart distribution with these parameters by $CW(K, N; M)$.

The following theorem is stated by Capon and Goodman [4]. It will be required repeatedly in the reduction process used to find the distribution of $\rho(\hat{M})$.

Theorem: If $N \times N$ matrix A is $CW(K, N; M)$, i.e., complex Wishart distribution with parameters K, N and covariance matrix M, and C is a nonsingular $N \times N$ matrix, then $B = CAC^*$ is $CW(K, N; CMC^*)$.

One wishes to find the distribution of the normalized signal-to-noise ratio in (16), i.e.,

$$\rho = (S^* \hat{M}^{-1} S)^2 / [(S^* M^{-1} S)(S^* \hat{M}^{-1} M \hat{M}^{-1} S)] \tag{24}$$

where $\hat{M}$ is the sample average estimate (12) of the covariance matrix M. To accomplish this, follow Capon and Goodman [4] by making a sequence of transformations which leave the random variable ρ in (16) fixed.

First, let $A = M^{-1/2} S$. Then, by (16),

$$\rho = (A^* \hat{M}_1^{-1} A)^2 / [(A^* A)(A^* \hat{M}_1^{-1} \hat{M}_1^{-1} A)] \tag{25}$$

where

$$\hat{M}_1 = M^{-1/2} \hat{M} M^{-1/2}.$$

The distribution of $\hat{M}$ is given by the complex Wishart distribution, defined in (22). That is, $\hat{M}$ is $CW(K, N; M)$ complex Wishart distributed with parameters K, N and covariance matrix M. By (25),

$$E\hat{M}_1 = M^{-1/2} (E\hat{M}) M^{-1/2} = M^{-1/2} M M^{-1/2} = I$$

where I denotes the identity matrix. If, in the above theorem, one lets A be $\hat{M}$ and C be $M^{-1/2}$, then B must be $\hat{M}_1$ and, by the theorem, $\hat{M}_1$ is $CW(K, N; I)$, where I is the $N \times N$ identity matrix.

Next, apply a transformation to (25) which normalizes vector A. That is, let

$$B = (A^* A)^{-1/2} A.$$

Then, ρ becomes

$$\rho = (B^* \hat{M}_1^{-1} B)^2 / B^* \hat{M}_1^{-1} \hat{M}_1^{-1} B \tag{26}$$

where $B^* B = 1$.

The above two transformations on ρ are the same that Capon and Goodman use in [4]. Following them further, use the fact that $B^* B = 1$ to construct an $N \times N$ unitary matrix U (a "rotation" in N-dimensional unitary space) which sends B into the column vector

$$P = (1, 0, 0, \cdots, 0)^*. \tag{27}$$

In other words, since B is a unit vector, there exists a unitary transformation U such that

$$B = UP \tag{28}$$

where P is the unit vector in (27). The application of transformation (28) to (26) yields

$$\begin{aligned} \rho &= [(UP)^* \hat{M}_1^{-1} (UP)]^2 / [(UP)^* \hat{M}_1^{-1} UU^* \hat{M}_1^{-1} (UP)] \\ &= [P^* (U^* \hat{M}_1 U)^{-1} P]^2 / [P^* (U^* \hat{M}_1 U)^{-1} (U^* \hat{M}_1 U)^{-1} P] \\ &= [P^* C^{-1} P]^2 / P^* C^{-2} P \end{aligned} \tag{29}$$

where

$$C = U^* \hat{M}_1 U \tag{30}$$

for the normalized signal-to-noise ratio ρ. By (22),

$$EC = U^* E\hat{M}_1 U = U^* U = I.$$

Thus, by the above theorem, C is $CW(K, N; I)$ distributed.

As suggested in [4], partition matrixes C, C^{-1} as

$$C = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}, \quad C^{-1} = \begin{pmatrix} C^{11} & C^{12} \\ C^{21} & C^{22} \end{pmatrix} \tag{31}$$

where C_{11} and C^{11} are scalars (1×1 matrixes).

Using the fact that $CC^{-1} = I$, this expression for ρ can be expressed as

$$\rho = 1 / [1 + (C_{12} C_{22}^{-1})(C_{12} C_{22}^{-1})^*] \tag{32}$$

in terms of only the matrixes C_{22} and C_{12} of the partition (31) of matrix C.

To find the distribution of ρ, as given in (32), start with the distribution of C. Since C is $CW(K, N; I)$ distributed, by (22), the joint probability density of the element of C is

$$P(C) = (|C|^{K-N} / I(I_N)) \exp [-\operatorname{tr}(C)] \tag{33}$$

where I_N is the N-dimensional identity matrix and $I(I_N)$ is the normalization constant given by (23). By a theorem of Frobenius, the partitioned matrix C can be factored as

$$C=\begin{pmatrix}1 & C_{12}\\ 0 & C_{22}\end{pmatrix}\begin{pmatrix}C_{11}-C_{12} & C_{22}^{-1}C_{21}, & 0\\ C_{22}^{-1}C_{21} & & , I\end{pmatrix}.$$

Hence, by Laplace's rules for computing determinants,

$$|C|=|C_{22}|(C_{11}-C_{12}C_{22}^{-1}C_{12}^*). \tag{34}$$

Using (34) in (33) and the fact that

$$\operatorname{tr} C = C_{11} + \operatorname{tr} C_{22},$$

one gets

$$P(C)=(|C_{22}|^{K-N}/I(I_N))(C_{11}-C_{12}C_{22}^{-1}C_{12}^*)^{K-N}$$
$$\cdot \exp[-(C_{11}-C_{12}C_{22}^{-1}C_{12}^*)]$$
$$\cdot \exp[-(C_{12}C_{22}^{-1}C_{12}^*+\operatorname{tr} C_{22})]. \tag{35}$$

Equation (35) corresponds to (17) in the paper by Capon and Goodman [4]. Following them, next make the change of variables

$$D_{11}=C_{11}-C_{12}C_{22}^{-1}C_{12}^*$$
$$D_{12}=C_{12}$$
$$D_{22}=C_{22} \tag{36}$$

in (35). Since the Jacobian of this transformation is one,

$$P(D_{11},D_{12},D_{22})=(|D_{22}|^{K-N}/I(I_K))D_{11}^{K-N}$$
$$\cdot \exp[-D_{11}-D_{12}D_{22}^{-1}D_{12}^*$$
$$-\operatorname{tr} D_{22}]$$
$$=P(D_{11})P(D_{12},D_{22}) \tag{37}$$

where

$$P(D_{11})\,K_1D_{11}^{K-N}\exp(-D_{11}),$$

a chi-square distribution with $2(K-N+1)$ degrees of freedom [4, eq. (19)], and

$$P(D_{12},D_{22})=K_2|D_{22}|^{K-N}\exp[-(D_{12}D_{22}^{-1})D_{12}^*$$
$$+\operatorname{tr} D_{22}] \tag{38}$$

where K_1 and K_2 are normalization constants. Substituting (36) in (32),

$$\rho = 1/[1+(D_{12}D_{22}^{-1})(D_{12}D_{22}^{-1})^*] \tag{39}$$

is the normalized signal-to-noise ratio in terms of the $N-1$-dimensional vector $D_{12}D_{22}^{-1}$. Next, in (38), let

$$E_{12}=D_{12}D_{22}^{-1}$$
$$E_{22}=D_{22}. \tag{40}$$

By the special form of this transformation, the Jacobian reduces to

$$\partial(E_{12},E_{22})/\partial(D_{12},D_{22}) = \begin{vmatrix}\partial E_{12}/\partial D_{12} & \partial E_{12}/\partial D_{22}\\ 0 & I\end{vmatrix}$$
$$=|\partial E_{12}/\partial D_{12}| = \partial(E_{12})/\partial(D_{12})$$

where $\partial E_{12}/\partial D_{12}$, etc., denotes, symbolically, the array or matrix partial derivatives associated with the transformation from D_{12} to E_{12}, etc. E_{12}, D_{12}, and D_{22}^{-1} are expressible in real and imaginary parts as

$$E_{12}=U+iV$$
$$D_{12}=X+iY$$
$$D_{22}^{-1}=F+iG.$$

Then,

$$E_{12}=D_{12}D_{22}^{-1}=XF-YG+i(XG+YF).$$

Hence,

$$U=XF-YG$$
$$V=XG+YF,$$

so that

$$\partial(E_{12})/\partial(D_{12})=\partial(U,V)/\partial(X,Y)=\begin{vmatrix}F & -G\\ G & F\end{vmatrix}$$
$$=|F+iG|^2=|E_{22}|^{-2}.$$

Thus, if one applies transformation (40) to (38), then

$$P(E_{12},E_{22})=K_2J^{-1}[(E_{12},E_{22})/(D_{12},D_{22})]\,|E_{22}|^{K-N}$$
$$\cdot \exp\{-[E_{12}(E_{22}E_{12})^*+\operatorname{tr} E_{22}]\}$$
$$=K_2|E_{22}|^{K-N+2}$$
$$\cdot \exp[-\operatorname{tr}(I+E_{12}^*E_{12})E_{22}] \tag{41}$$

is the joint probability density of the components of E_{12} and E_{22}. The last step in (41) follows from the fact that $E_{12}(E_{22}E_{12})^*$ is scalar, and, as a consequence,

$$E_{12}(E_{22}E_{12})^* = \operatorname{tr}(E_{12}^*E_{12}E_{22}).$$

Observe that the right side of (41) is, if matrix $I + E_{12}{}^*E_{12}$ is held constant, of the same form as the complex Wishart distribution developed by Goodman [3]. In fact, it has the form the distribution would have if it were $CW(K+1, N-1; [I + E_{12}{}^*E_{12}]^{-1})$ distributed. By (22), such a complex Wishart distribution of E_{12} would be

$$P(E_{22}) = K_3 \,|E_{22}|^{K-N+2} |I + E_{12}{}^*E_{12}|^{K+1}$$
$$\cdot \exp\left[-\operatorname{tr}\left\{I + E_{12}{}^*E_{12}\right\} E_{22}\right]$$

where K_3 is the normalizing constant. Hence, (41) has the form

$$P(E_{12}, E_{22}) = K_3 [1/\,|I + E_{12}{}^*E_{12}|^{K+1}] P(E_{22}).$$

Since the integral of $P(E_{22})$ over domain $D_{E_{22}}$ is unity (see [4, Theorem 5.1]),

$$P(E_{12}) = K_4 [1/\,|I + E_{12}{}^*E_{12}\,|^{K+1}] \tag{42}$$

where K_3 is the constant of normalization.

The joint probability density of

$$E_{12} = D_{12}D_{22}{}^{-1} = C_{12}C_{22}{}^{-1},$$

as given by (42), can be further reduced in complexity by observing that $E_{12}{}^*E_{12}$ is a Hermitian matrix of rank one. Since the determinant of a Hermitian matrix equals the product of its eigenvalues,

$$|I + E_{12}{}^*E_{12}| = 1 + E_{12}E_{12}{}^*$$

and, by (42), the joint probability density of the elements of E_{12} is now given by

$$P(E_{12}) = K_4 [1/(1 + E_{12}E_{12}{}^*)^{K+1}]. \tag{43}$$

Since E_{12} in (43) is a row vector of $N-1$ complex numbers, the "inner" product $E_{12}E_{12}{}^*$ can be written as a sum of squares. Changing to polar coordinates in $2N-2$ dimensions, (43) with its associated volume element is

$$P(E_{12})dV = K_3 \cdot [1/(1 + r^2)^{K+1}] r^{2N-3} \, dr d\Omega \tag{44}$$

where

$$r^2 = E_{12}E_{12}{}^*$$

and $d\Omega$ is a differential of solid angle in $2N-2$ dimensions. Integrating over all solid angles yields the density

$$P(r) = K_4 [r^{2N-3}/(1 + r^2)^{K+1}] \tag{45}$$

of the "radial" coordinate, where constant K_4 is K_3 times the surface "area" of the unit "sphere" in $2N-2$ dimensions.

In terms of r, the normalized signal-to-noise ratio ρ in (32) is given by

$$\rho = 1/(1 + r^2). \tag{46}$$

Changing variables in (45) from r to ρ yields

$$P(\rho) = K_5 (1-\rho)^{N-2} \rho^{K+1-N} \tag{47}$$

where $0 \leqslant \rho \leqslant 1$ is the probability density of ρ. The normalization constant K_5 in (47) is obtained from the well-known integral

$$\int_0^1 t^m (1-t)^n \, dt = n!m!/(m+n+1)!$$

where n and m are integers greater than -1.

Using this in (47) yields, finally, the desired probability distribution of ρ, due to Goodman, namely,

$$\operatorname{Prob}\left\{\rho \leqslant X\right\} = \left\{K!/[(N-2)!(K+1-N)!]\right\}$$
$$\cdot \int_0^X (1-\rho)^{N-2} \rho^{K+1-N} \, d\rho \tag{48}$$

where $0 \leqslant X \leqslant 1$. The probability distribution of the normalized signal-to-noise ratio ρ is an incomplete beta function distribution. From this fact, the first and second moments of this distribution are

$$E\rho = (K + 2 - N)/(K + 1)$$

and

$$E\rho^2 = [(K + 2 - N)(K + 3 - N)] / [(K + 2)(K + 1)]$$

This result for the first moment is used to define the expected loss in power ratio, given in (19).

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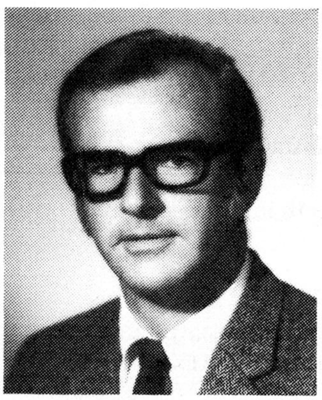

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